

Business Economic and Financial Data

Analysis of Player Numbers Fluctuation Using Time Series Models

Shiyu Min
2141876

Table of Contents

<i>Background.....</i>	<i>4</i>
<i>Data Description</i>	<i>4</i>
<i>Methodology.....</i>	<i>4</i>
<i>Analysis</i>	<i>6</i>
<i>Baldur's Gate 3</i>	<i>6</i>
<i>Elden Ring</i>	<i>8</i>
<i>Sid Meier's Civilization VI</i>	<i>11</i>
<i>Discussion</i>	<i>14</i>
<i>Conclusion</i>	<i>14</i>
<i>Bibliography.....</i>	<i>16</i>
<i>Codes</i>	<i>17</i>

Background

The video game industry is characterized by rapid growth and intense competition. Game developers employ a variety of strategies to attract and retain players while adapting to changing market dynamics and evolving consumer preferences. This report examines three popular games from different genres, each representing a distinct development strategy that influences its market success and player retention.

Data Description

This report utilizes datasets from three video games downloaded from SteamDB. the data is about daily peak player counts, including a mix if both new and returning players. Based on official reports from each game's publisher regarding unique copies sold, have estimated the daily new player counts. This estimate is derived from the ratio of unique copies to the peak daily player numbers. Although it is approximate, this method that the distribution of daily peak numbers is closed to the unique copies numbers in developer given records.

Methodology

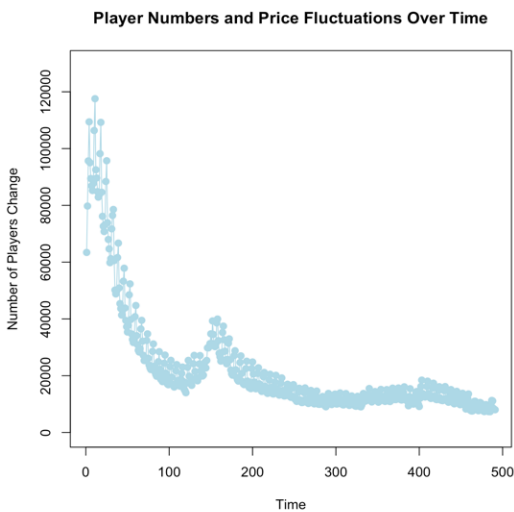


Figure 1.1

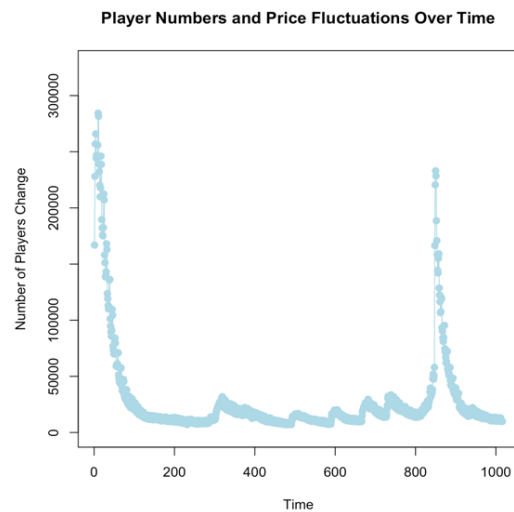


Figure 1.2

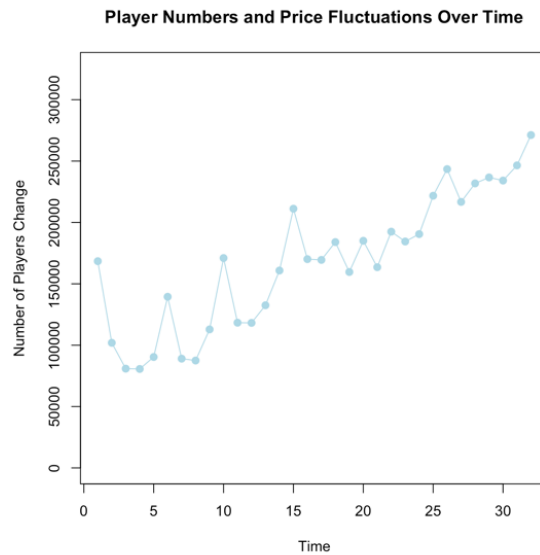


Figure 1.3

Figure 1.1 The player number fluctuation of Baldur's Gate 3 daily. Figure 1.2 The player number fluctuation of Elden Ring daily. Figure 1.3 The player number fluctuation of Sid Meier's Civilization quarterly.

Linear models seem works not well for these three datasets. Both the Bass model and the Arima model are more suitable for analysis. Additionally, the Prophet model is applied to do forecast the datasets of Baldur's Gate 3 and Elden Ring.

Figure 1.1 is the plot of the player number for Baldur's Gate 3, characterized by an extreme initial peak followed by a smaller peak. The general Bass model is not appropriate for this dataset, employing a Generating Bass model with two exponential shocks provides insightful results. Using trend method or exponential smoothing method are not ideal due to the rapid decrease in numbers post peak, whereas the Prophet performs well, particularly incorporating shocks into the model.

Figure 1.2 shows the player data for Elden Ring. It uses the same GBM for analysis and the Prophet model for forecasting as used in BG3. However, the plot has two shocks spaced over a long interval. In the GBM analysis process, splitting the dataset into two parts gets a more accurate analysis.

Figure 1.3 features Sid Meier's Civilization VI, which differs from figure 1.1 and 1.2, because the game was released much earlier, about nine years ago. Analyzing daily data can involve many fluctuations; by converting the daily changes in player numbers to quarterly data using Python, the plot exhibits distinct characteristics

comparing with other two. The general BM provides a good fit for this dataset, the ARIMA model is used for additional insights. Ultimately, a more sophisticated SARIMAX model is applied for forecasting.

Analysis

Baldur's Gate 3

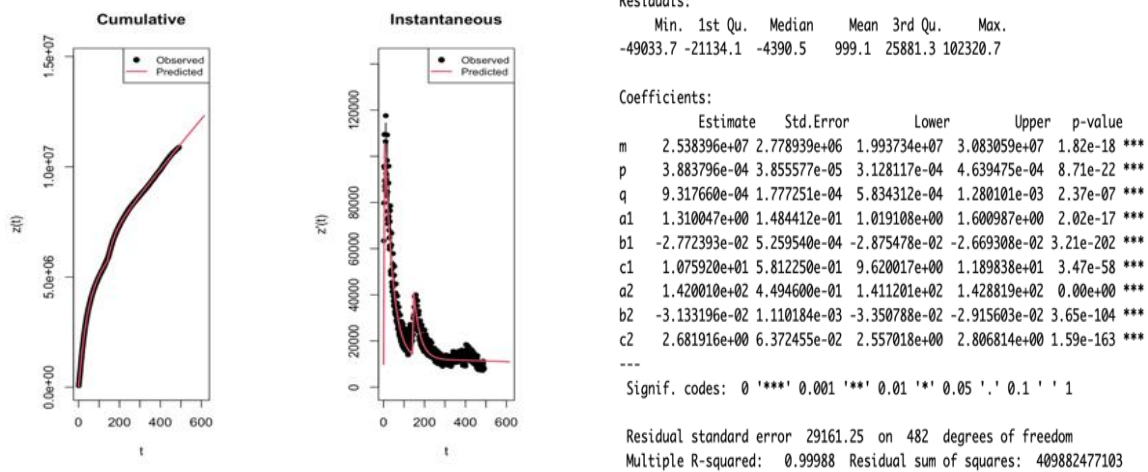


Figure 2.1

The figure presents a plot of Baldur's Gate 3 player data using a GBM with two exponential shocks. The cumulative plot and its predictions indicate a growing trend, suggesting that the total number of players will continue to increase in the future. The instantaneous graph displays a sharp peak at the initial stage of the game, corresponding to its release on August 3, 2023, coinciding with Larian Studios officially ending early access. Ten days later, on August 13, 2023, the daily number of players reached its maximum.

The $p = 3.883796e-04$ and $q = 9.317660e-04$ indicates that the internal influence imitation is higher than the external influence innovation.

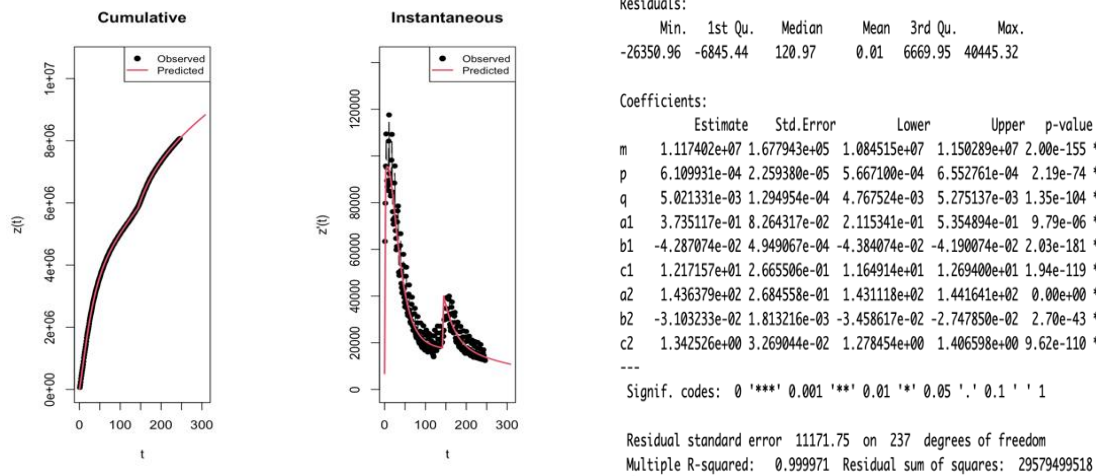


Figure 2.2

The figure provides a detailed analysis of half the dataset. It reveals that the coefficient of imitation, $q=5.021331e-03$ is notably higher than the coefficient of innovation, $p=6.109931e-04$. This disparity suggests that existing players' influence on attracting new players is stronger than external marketing influences.

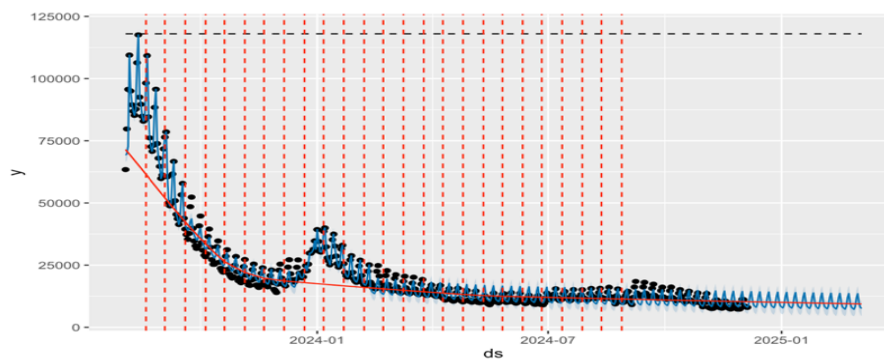


Figure 2.3

This figure was generated using an additive Prophet model, which automatically identified seasonality as daily and incorporated 25 changepoints to accommodate shifts in the trend. The forecast extends three months after the last actual data point, illustrating daily fluctuations and predicting a near flat trend continuation.

The model marked 25 change points, in the aspect of trend shows exponentially decreasing, in the last 20% of the dataset and extending into the future prediction, the trend tends to flatten, indicating a stabilization in player numbers.

Elden Ring

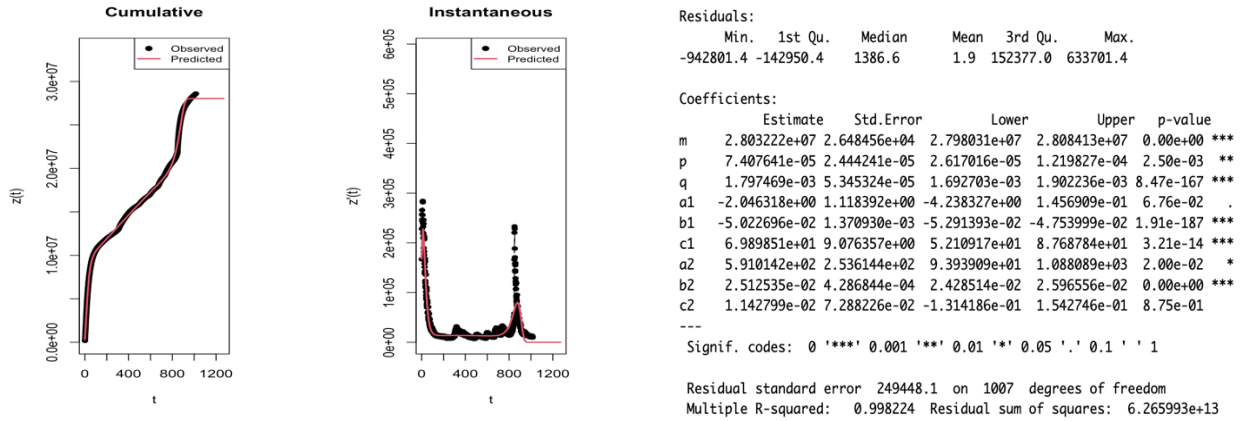


Figure 2.4

The instantaneous plot for Elden Ring displays two obvious peaks. The first one is on 5 Mar 2022, ten days after its full releasing, the daily players number reached 953,426 in average. The second peak is on 22 Jun 2024, coincided with its new DLC releasing on 21 Jun 2024, before this event happened, the players number had a significant increasing started from 19 Jun 2024 with 194,570, on 20 Jun 2024 with 558,139, finally on 22 Jun 2024, it reached the maximum 781,261. Its cumulative plot reflects the data performance. In the beginning stage, showing a rapid increase in total player numbers, followed by a period of stable growth, and it suddenly increased after the new DLC released.

In summary, its p and q are 7.407641e-05 and 1.797469e-03 respectively, indicating that imitation has more important influence than innovation, but considering the game had substantial changes during the dataset's lifecycle, requiring more detailed analysis.

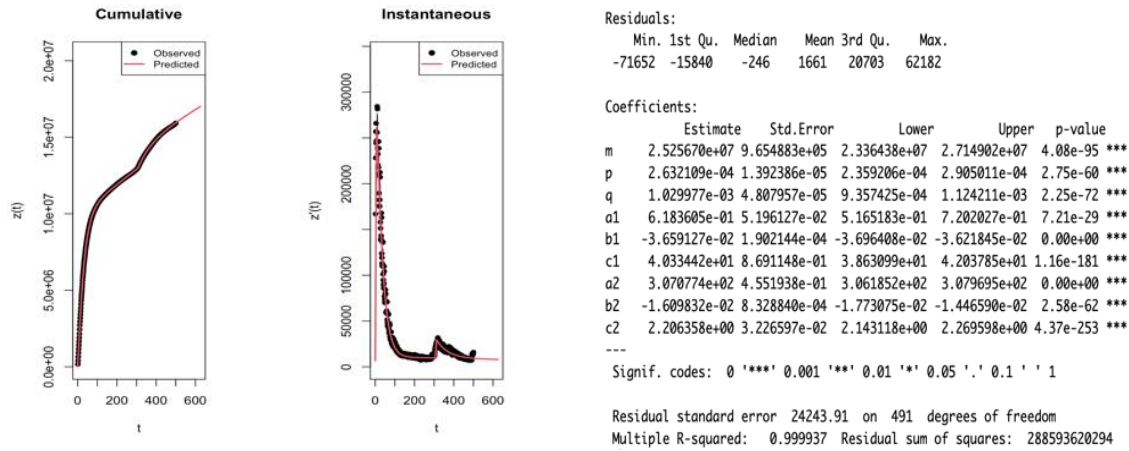


Figure 2.5

The plot zooms on half of the dataset, the gap between imitation and innovation effects is smaller, although social influence imitation continues to play a more important role than innovation, both coefficients can impact player numbers.

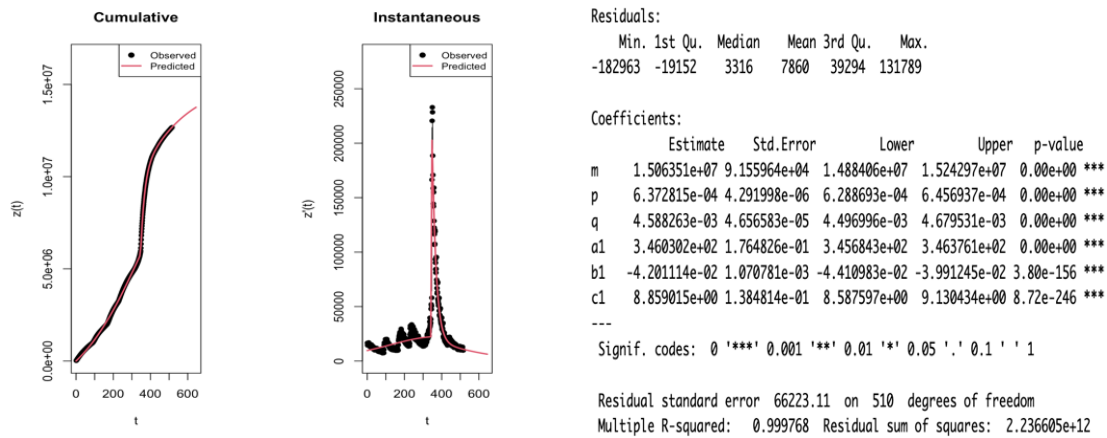


Figure 2.6

In the latter half of the dataset, the coefficient of imitation remains higher than that of innovation, and their gap is also narrow.

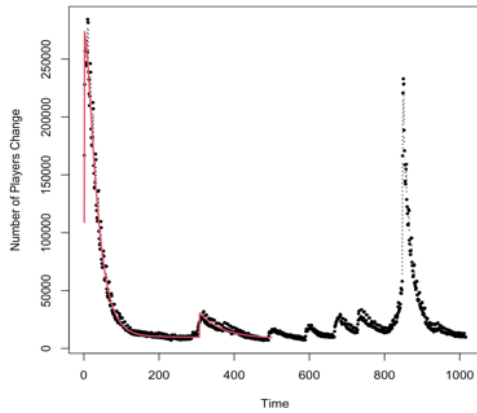


Figure 2.7

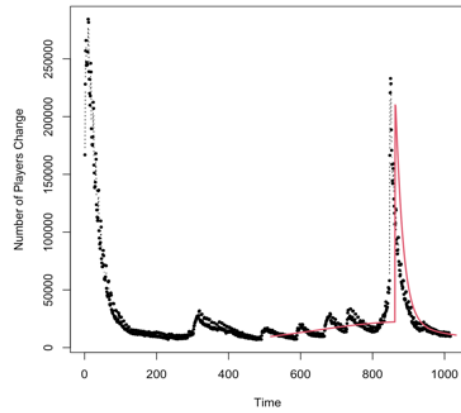


Figure 2.8

The prediction model accurately fits the first half of the data, capturing two significant peaks. However, the second half of the data shows several fluctuations before reaching the peak, which presents challenges for the model. Despite these difficulties, the model represents an increasing trend that closely predicts the overall tendency of player numbers. There are several fluctuations in mid stage of the dataset, due to the platform Steam's promotion and seasonal variations of price, which can improve a video game's sale amount.

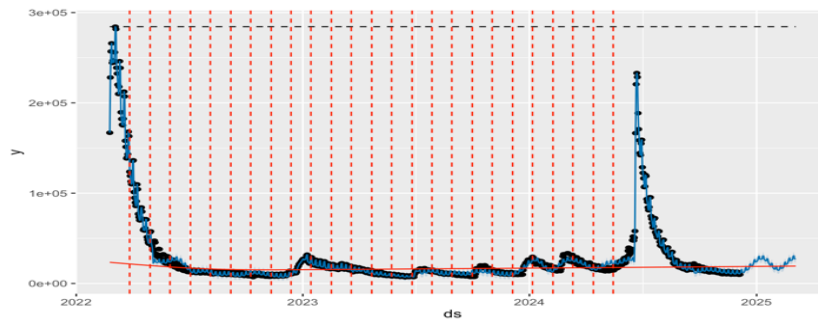


Figure 3.9

Based on historical trends observation, player numbers are expected to experience minor fluctuations from December 2024 to February 2025, assuming no major updates are released by the game company during this period.

Sid Meier's Civilization VI

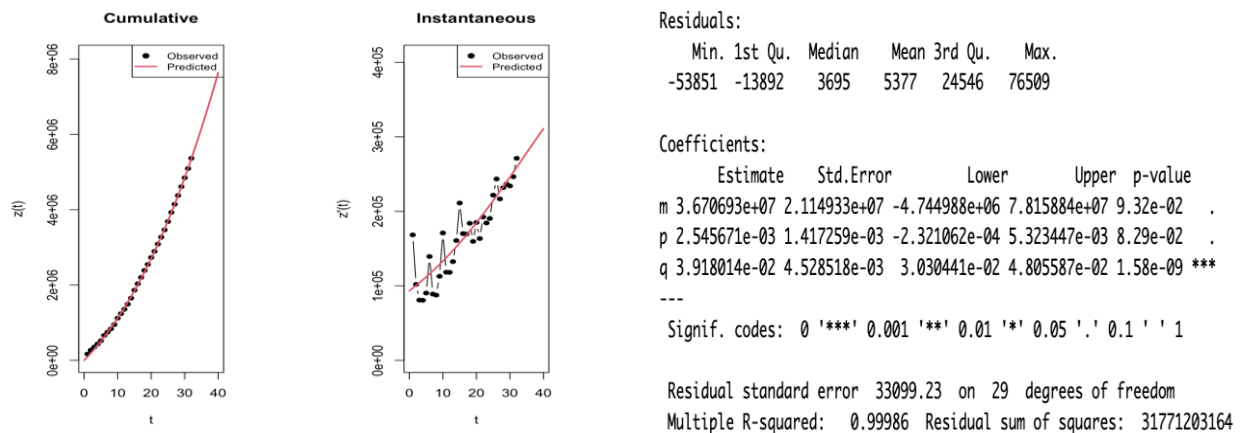


Figure 2.10

The plot is generated using the Bass Model, and unlike the previous two datasets, this time series has been converted to quarterly data. The innovation $p = 2.545671e-03$ and imitation $q = 3.918014e-02$ with R-squared 0.99986, indicating that fluctuations in player numbers are primarily influenced by the spread among existing players and community. Although the Bass Model effectively predicts the general trend of the dataset, several outliers in the instantaneous plot do not align with the model's predictions.

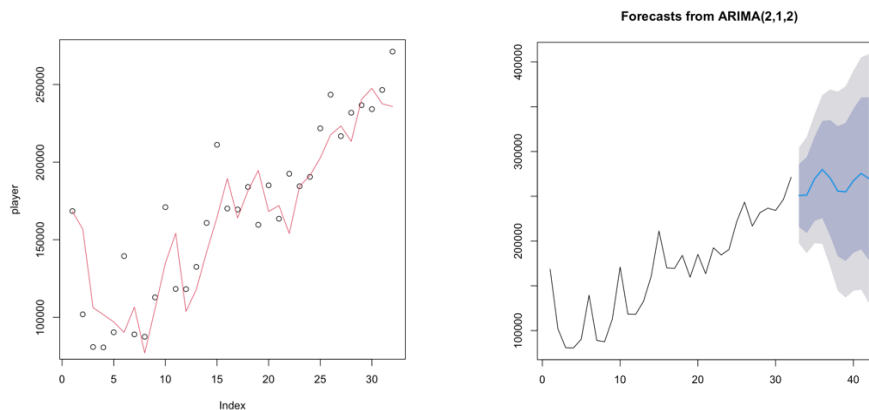


Figure 2.11

Figure 2.12

The figures represent the ARIMA model fit and forecast for player data. The model closely follows most data points, attempting to capture significant events within the dataset. Each point is the player counts quarterly, there are three points are

outliers the red line does not approach exactly, this substantial changes because of the game developer Firaxis Games released three main DLC on the specific dates during these periods.

The ARIMA(2,1,2) forecast suggests that the player count will continue to trend upwards over the next four quarters.

However, there are certain dynamics that a single model may not fully capture, indicating the need for further analysis. Prior to this, it would be prudent to perform additional tests on the dataset

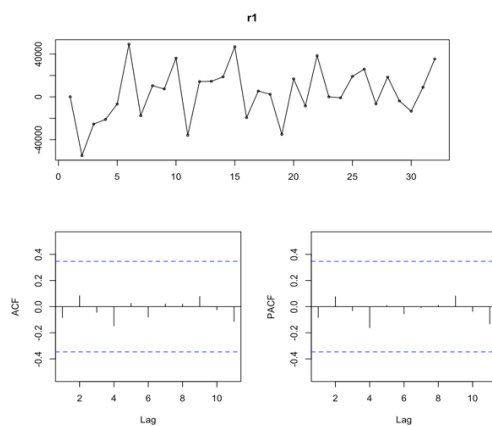


Figure 2.13

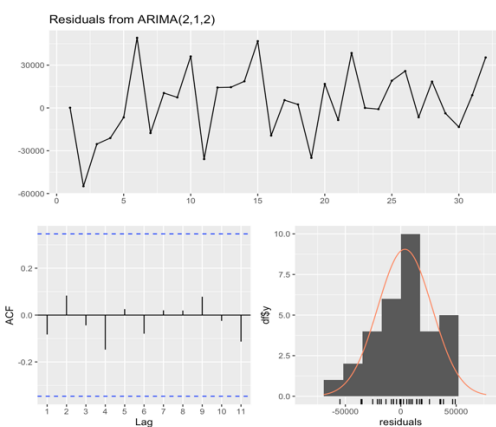


Figure 2.14

In residual plots, the time series do not show obvious pattern for seasonality and trend, meaning the residuals may represent white noise exist. The lags in both ACF and PACF are contained within the confident interval indicates there are no significant autocorrelations, so the model is appropriate for this dataset.

In the check residuals model, the histogram and density plot show that the residuals are approximately distributing, but there is slight skewness existing. Furthermore, the Ljung-Box Test results in a p-value = 0.6333, it is a strong evidence fail to reject the null hypothesis.

Series: cumsum(player)
 Regression with ARIMA(2,1,2)(1,0,1)[4] errors

Coefficients:

	ar1	ar2	ma1	ma2	sar1	sma1	xreg
	-0.1026	-0.9068	0.3178	1.0000	-0.0374	0.0406	0.9961
s.e.	0.1152	0.1000	0.1216	0.1545	NaN	NaN	0.0210

sigma^2 = 409273344: log likelihood = -349.05
 AIC=714.1 AICc=720.65 BIC=725.58

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	-991.9274	17520.13	13947.28	-0.4102198	1.210877	0.08320614	-0.04071275

Utilize SARMAX to do further analysis, its summary with several crucial features coefficients and training set error measures. Overall, the model captures the dataset's features well, but there are some issues. High RMSE and MAE indicate the model capturing the pattern with large errors.

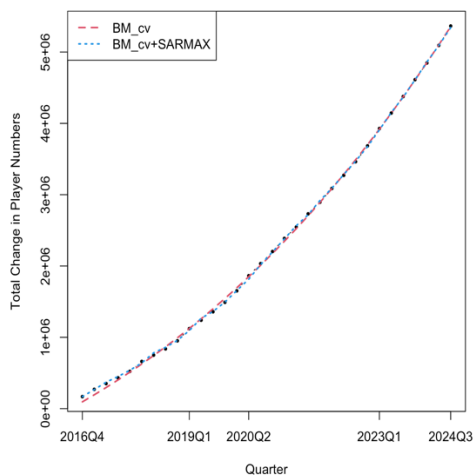


Figure 2.15

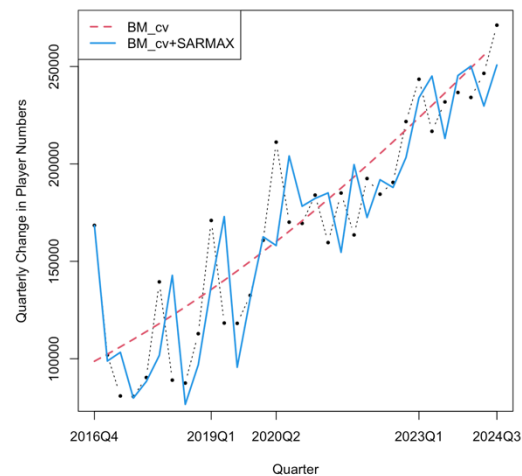


Figure 2.16

In the cumulative plot, the BM and its incorporating with SARMAX predicts a similar trend, and there is no obvious difference of capturing the dynamic factors between two models, BM works well in cumulative plot.

In the instantaneous plot, the blue line follows the point closer than the red line and represent an upward tendency. The enhanced model can predict the data well

comparing with BM, the players numbers fluctuation is dynamic growing with the time, BM incorporates SARMAX adheres each point and outliers.

Discussion

The name of game 'Baldur's Gate' is a popular factor among RPG fans, and Larian studio released the game's early access, allowing the players buy and playing the parts of the game that already have been developed before its full release. This approach gave the feedback to the game, more importantly players shared their experience and drawing new players.

The development company of Elden Ring FromSoftware is a well-known company in video game industry, its games with a distinctive style, noted as Souls-like, the EDR also continues to this tradition, that is one of one reason its imitation is always higher, but its innovation can reflect FromSoftware's new ideas for this game. The author of the "A song of ice and fire" series by George R.R. Martin crafted the game's lore and world. Moreover, EDR is the first time of true open world game for FromSoftware comparing with their earlier products. EDR's original and new features seem help FromSoftware achieve good effect, so the player grew dramatically when the new DLC released.

Sid Mier's Civilization has been released approximate nine years, its players number is still growing in time, this trend can represent a success of the developer Firaxis Games, although it is not unique in video game industry. There are many reasons can explain CV6's growing trend, two of them might be the most crucial factors. a robust community that actively contributes new mods and content, keeping the game vibrant and fresh, and the other reason is the game's appeal to fans of historical and strategy genres.

Conclusion

This report demonstrates three games with different categories and background, highlighting how developer strategies such as early access for "Baldur's Gate 3" (BG-3), DLC releases for "Elden Ring" (EDR), and robust community engagement for "Sid Meier's Civilization VI" (CV-6) have significantly influenced their

popularity and player dynamics. Although this report did not involve the topic about direct or undirect competition from other games, it is evident appeared factor, the interactions between players play a crucial role in shaping game adoption. The significance of the imitation coefficient in our models underscores the impact of existing players in attracting new ones, emphasizing the importance of community and shared players' experiences in driving game popularity.

Bibliography

Andric, D. (2023, February 3). *How many copies did Civilization sell?* Retrieved from levvel: <https://levvel.com/civilization-statistics/>

Bevan , R. (2024, november 25). *Larian Reported A Loss Of €214,000 Before Baldur's Gate 3's Launch*. Retrieved from thegamer: <https://www.thegamer.com/baldurs-gate-3-sales-15-million-larian-losses/#:~:text=Summary,to%20Baldur's%20Gate%203's%20launch.>

Chavez, T. (2024, December 15). *Elden Ring Has Sold Over 28.6 Million Copies As FromSoftware Shatters Sales Records*. Retrieved from 9meters: <https://9meters.com/entertainment/games/elden-ring-has-sold-28-6-million-copies#:~:text=Elden%20Ring%20has%20sold%20over%2028.6%20million%20copies%20worldwide%20as,status%20as%20a%20breakout%20hit.>

Guidolin, M. (2024). *Innovation Diffusion Models Theory and Practice*: Wiley.

Codes

Baldur's Gate 3

```
library(readxl)
library(prophet)
library(DIMORA)
library(forecast)

BG3<- read_excel("Baldur's Gate 3 _Players.xls")
Players = BG3$Players * 0.1343096
player <- ts(Players, frequency = 1, start = 1)

BG3_price<-read_excel("Baldur's Gate 3 _Price.xlsx")
price_change <- BG3_price$`Final price`

par(mar=c(5, 4, 4, 6) + 0.1) # Adjust the last value for the right margin as needed

# Plot for player changes
plot(player, type="o", col="#ADD8E6", xlab="Time", ylab="Number of Players Change",
      main="Player Numbers and Price Fluctuations Over Time", pch=16, cex=1.2, lty=1,
      ylim=c(0, max(player)*1.1))

par(new=TRUE)
plot(price_change, type="o", col="#F08080", axes=FALSE, xlab="", ylab="", pch=17, cex=1.2,
      lty=2, ylim=c(0, max(price_change)*1.1))
axis(side=4, col="#F08080", col.axis="#F08080")
mtext("Price Fluctuation", side=4, line=3, col="#F08080")

legend("topright", inset=c(-0.004,0), legend=c("Number of Players Change", "Price
Fluctuation (€)"),
      col=c("#ADD8E6", "#F08080"), lty=1:2, pch=16:17, cex=0.8, pt.cex=1.5, bty="n",
      xpd=TRUE)

###prediction
bm_bg <- GBM(player,shock = "exp",nshock = 2, prelmestimates = c(1.088884e+07, 0.003 ,
0.1, 11, -0.9, 1, 152, -0.5, 1))
summary(bm_bg)

###prediction
pred_bmbg<- predict(bm_bg, newx=c(1:492))
```



```

pred.instbg<- make.instantaneous(pred_bmbg)

###plot of fitted model
plot(cumsum(player), type= "b",xlab="Time", ylab="Total Change in Player Numbers",
pch=16, lty=1, cex=0.8, xlim=c(1,492), ylim=c(0,1.2e+7))
lines(pred_bmbg, lwd=2, col=2)

plot(player, type= "b",xlab="Time", ylab="Number of Players Change", pch=16, lty=3,
cex=0.6, xlim=c(1,492))
lines(pred.instbg, lwd=2, col=2)

###estimate the model with 50% of the data
bm_bg50<-GBM(player[1:246],shock = "exp",nshock = 2, prelmestimates =
c(1.088884e+07, 2.729707e-04 , 1.270825e-02, 11, -0.9, 1, 152, -3.892836e-02, -
8.554876e-01))
summary(bm_bg50)

pred_bmbg50<- predict(bm_bg50, newx=c(1:246))
pred.instbg50<- make.instantaneous(pred_bmbg50)

plot(player, type= "b",xlab="Time", ylab="Number of Players Change", pch=16, lty=3,
cex=0.6)
lines(pred.instbg50, lwd=2, col=2)

# Prophet
Average_player <- data.frame(player, BG3$DateTime, cap=118000)
colnames(Average_player)=c("y","ds","cap")
plot(Average_player$y, x=Average_player$ds, ylab = "Number of Players Change", xlab =
"Day")

shock_dates <- data.frame(
  holiday = c('shock_event_1', 'shock_event_2'),
  ds = as.Date(c('2023-08-13', '2024-01-07')), # Example shock dates
  lower_window = c(-7,-20),
  upper_window = c(35,35) # Duration of the effect
)

m = prophet(Average_player, growth = "logistic", holidays = shock_dates)
summary(m)

##create a future 'window' for prediction
future <- make_future_dataframe(m, periods = 90, freq="day", include_history = T)
tail(future)
future$cap=118000

```

```

forecast <- predict(m, future)
tail(forecast[c("ds", "yhat", "yhat_lower", "yhat_upper")])

plot(m, forecast)

#Dynamic plot
dyplot.prophet(m, forecast)

#plot with change points

plot(m, forecast)+add_changepoints_to_plot(m, threshold=0)

#dates corresponding to change points
m$changepoints

##Prophet with no change points and multiplicative seasonality
m2=prophet(Average_player, growth="logistic", n.changepoints=0, holidays = shock_dates,
seasonality.mode='multiplicative')

summary(m2)
m2$seasonalities

future2 <- make_future_dataframe(m2, periods = 90, freq="day", include_history = T)
tail(future2)
future2$cap=118000

forecast2 <- predict(m2, future2)
tail(forecast2[c("ds", "yhat", "yhat_lower", "yhat_upper")])

#prediction plot

plot(m2, forecast2)
dyplot.prophet(m2, forecast2)

```

Elden Ring

```
library(readxl)
library(prophet)
library(DIMORA)
library(forecast)
```

```
Edr<- read_excel("Eleden Ring Players.xls")
Players = Edr$Players * 0.2982192
player <- ts(Players, frequency = 1, start = 1)
```

```
Edr_price <- read_excel("EDR Price.xlsx")
price_change <- Edr_price$`Final price`
```

```
par(mar=c(5, 4, 4, 6) + 0.1) # Adjust the last value for the right margin as needed
```

```
# Plot for player changes
plot(player, type="o", col="#ADD8E6", xlab="Time", ylab="Number of Players Change",
      main="Player Numbers and Price Fluctuations Over Time", pch=16, cex=1.2, lty=1,
      ylim=c(0, max(player)*1.15))
```

```
par(new=TRUE)
plot(price_change, type="o", col="#F08080", axes=FALSE, xlab="", ylab="", pch=17, cex=1.2,
      lty=2, ylim=c(0, max(price_change)*1.15))
axis(side=4, col="#F08080", col.axis="#F08080")
mtext("Price Fluctuation", side=4, line=3, col="#F08080")
```

```
legend("topright", inset=c(-0.004,0), legend=c("Number of Players Change", "Price
Fluctuation (€)"),
      col=c("#ADD8E6", "#F08080"), lty=1:2, pch=16:17, cex=0.8, pt.cex=1.5, bty="n",
      xpd=TRUE)
```

```
###prediction
```

```
bm_edr <- GBM(player,shock = "exp",nshock = 2, prelmestimates = c(2.803218e+07,
7.386279e-05, 1.797903e-03, 10, -5.023506e-02, 6.997694e+01, 850, -2.512700e-02,
1.142238e-02))
summary(bm_edr)
```

```
###prediction
```

```
pred_bmedr<- predict(bm_edr, newx = (1:1016))
pred.instedr<- make.instantaneous(pred_bmedr)
```

```
###plot of fitted model
```

```
plot(cumsum(player), type= "b",xlab="Time", ylab="Total Change in Player Numbers",  
pch=16, lty=1, cex=0.8)  
lines(pred_bmedr, lwd=2, col=2)
```

```
plot(player, type= "b",xlab="Time", ylab="Number of Players Change", pch=16, lty=3,  
cex=0.6)  
lines(pred.instedr, lwd=2, col=2)
```

```
###estimate the model with 50% of the data  
bm_edr50<-GBM(player[1:500],shock = "exp",nshock = 2, prelmestimates =  
c(2.8600004e+07, 2.550158e-05, 8.935422e-04, 10, -3.027675e-02, 3.701211e+00, 319, -  
0.001, 0.1))  
summary(bm_edr50)
```

```
pred_bmedr50<- predict(bm_edr50, newx=c(1:500))  
pred.instedr50<- make.instantaneous(pred_bmedr50)
```

```
plot(player, type= "b",xlab="Time", ylab="Number of Players Change", pch=16, lty=3,  
cex=0.6)  
lines(pred.instedr50, lwd=2, col=2)
```

```
###estimate the model with 50% to 100% of the data  
bm_edr50_100<-GBM(player[501:1016],shock = "exp",nshock = 1, prelmestimates =  
c(2.8600004e+07, 6.389492e-04, 4.552686e-03, 350, -4.142344e-02, 8.827355e+00))  
summary(bm_edr50_100)
```

```
pred_bmedr50_100<- predict(bm_edr50_100, newx=c(1:516))  
pred.instedr50_100<- make.instantaneous(pred_bmedr50_100)
```

```
x_values <- 1:length(pred.instedr50_100)  
adjusted_x <- x_values + 516  
plot(player, type= "b", xlab="Time", ylab="Number of Players Change", pch=16, lty=3,  
cex=0.6)  
lines(adjusted_x, pred.instedr50_100, lwd=2, col=2)
```

```
# Prophet  
Average_player <- data.frame(player, Edr$DateTime, cap=284330)  
colnames(Average_player)=c("y","ds","cap")  
plot(y = Average_player$y, x = Average_player$ds, type = "o",  
xlab = "Day", ylab = "Number of Players Change", lwd = 2, pch = 16, cex = 1.2)
```

```
shock_dates <- data.frame(  
holiday = c('shock_event_1', 'shock_event_2', 'shock_event_3'),  
ds = as.Date(c('2022-03-05', '2023-01-08', '2024-06-22')), # Example shock dates
```

```

lower_window = c(-10,-18,-3),
upper_window = c(60,180,90) # Duration of the effect
)

m = prophet(Average_player, growth = "logistic", holidays = shock_dates)
summary(m)

future <- make_future_dataframe(m, periods = 90, freq="day", include_history = T)
tail(future)
future$cap=284330

forecast <- predict(m, future)
tail(forecast[c("ds", "yhat", "yhat_lower", "yhat_upper")])

plot(m, forecast)

#Dynamic plot
dyplot.prophet(m, forecast)

#plot with change points

plot(m, forecast) +add_changepoints_to_plot(m, threshold=0)

#dates corresponding to change points
m$changepoints

##Prophet with no change points and multiplicative seasonality
m2=prophet(Average_player, growth="logistic", n.changepoints=0, holidays = shock_dates,
seasonality.mode='multiplicative')

summary(m2)
m2$seasonalities

future2 <- make_future_dataframe(m2, periods = 90, freq="day", include_history = T)
tail(future2)
future2$cap=284330

forecast2 <- predict(m2, future2)
tail(forecast2[c("ds", "yhat", "yhat_lower", "yhat_upper")])

#prediction plot

plot(m2, forecast2)
dyplot.prophet(m2, forecast2)

```

Sid Meier's Civilization

```
library(readxl)
library(DIMORA)
library(forecast)
library(tseries)

cv<- read_excel("Sid Meier's Civilization VI.xlsx")
Players = cv$Players * 0.04366329
player <- ts(Players, frequency = 1, start = 1)
player <- player[-length(player)]

price <- cv$`Final Price`
price <- price[-length(price)]

quarter <-cv$Quarter
quarter <- quarter[-length(quarter)]

par(mar=c(5, 4, 4, 6) + 0.1) # Adjust the last value for the right margin as needed

# Plot for player changes
plot(player, type="o", col="#ADD8E6", xlab="Time", ylab="Number of Players Change",
      main="Player Numbers and Price Fluctuations Over Time", pch=16, cex=1.2, lty=1,
      ylim=c(0, max(player)*1.2))

par(new=TRUE)
plot(price, type="o", col="#F08080", axes=FALSE, xlab="", ylab="", pch=17, cex=1.2, lty=2,
      ylim=c(0, max(price)*1.2))
axis(side=4, col="#F08080", col.axis="#F08080")
mtext("Price Fluctuation", side=4, line=3, col="#F08080")

legend("topright", inset=c(-0.004,0), legend=c("Number of Players Change", "Price
Fluctuation (€)"),
      col=c("#ADD8E6", "#F08080"), lty=1:2, pch=16:17, cex=0.8, pt.cex=1.5, bty="n",
      xpd=TRUE)

#Bass Model
bm_cv <- BM(player, display = TRUE)

summary(bm_cv)

pred_bmcv<- predict(bm_cv, newx=c(1:32))
pred.instcv<- make.instantaneous(pred_bmcv)
```

```
plot(cumsum(player), type= "b",xlab="Time", ylab="Total Change in Player Numbers",  
pch=16, lty=3, cex=0.6)  
lines(pred_bmcv, lwd=2, col=2)
```

```
plot(player, type= "b",xlab="Time", ylab="Quarterly Change in Player Numbers", pch=16,  
lty=3, cex=0.6)  
lines(pred.instcv, lwd=2, col=2)
```

```
####Arima model  
a1<- Arima(player, order=c(2,1,2), seasonal=c(1,0,1)[4])  
fit1<- fitted(a1)  
summary(fit1)#!
```

```
plot(player)  
lines(fit1, col=2)
```

```
f1<- forecast(a1)  
plot(f1)
```

```
r1<- residuals(a1)  
tsdisplay(r1)
```

```
checkresiduals(f1)
```

```
##Exploratory plots  
plot(player, type= "b",xlab="Quarter", ylab="Quarterly Change in Player Numbers", pch=16,  
lty=3, cex=0.6, xaxt="n")  
axis(1, at=c(1, 10, 15, 26, 32), labels=format(quarter[c(1, 10, 15, 26, 32)]))
```

```
plot(cumsum(player), type= "b",xlab="Quarter", ylab="Total Change in Player Numbers",  
pch=16, lty=3, cex=0.6, xaxt="n")  
axis(1, at=c(1, 10, 15, 26, 32), labels=format(quarter[c(1, 10, 15, 26, 32)]))
```

```
##BM  
bm_cv<- BM(player, display = TRUE)
```

```
##Prediction with pred_bm_cv2  
pred_bm_cv<- predict(bm_cv, newx=c(1:32))  
pred_bm_cv_i<- make.instantaneous(pred_bm_cv)[-1]
```

```
##Plots observed vs predicted  
plot(player, type= "b",xlab="Quarter", ylab="Quarterly Change in Player Numbers", pch=16,  
lty=3, cex=0.6, xaxt="n", col=1)  
axis(1, at=c(1, 10, 15, 26, 32), labels=format(quarter[c(1, 10, 15, 26, 32)]))
```

```
lines(pred_bm_cv_i, lwd=2, col=2)
```

```
plot(cumsum(player), type= "b",xlab="Quarter", ylab="Total Change in Player Numbers",  
pch=16, lty=3, cex=0.6, xaxt="n",col=1)  
axis(1, at=c(1, 10, 15, 26, 32), labels=format(quarter[c(1, 10, 15, 26, 32)]))  
lines(pred_bm_cv, lwd=2, col=2)
```

```
##SARMAX refinement
```

```
fit.player<- fitted(bm_cv)
```

```
s2 <- Arima(cumsum(player), order = c(2,1,2), seasonal=list(order=c(1,0,1), period=4), xreg  
= fit.player)
```

```
summary(s2)
```

```
pres2 <- make.instantaneous(fitted(s2))
```

```
##Plots observed vs predicted with SARMAX refinement
```

```
plot(player, type= "b",xlab="Quarter", ylab="Quarterly Change in Player Numbers", pch=16,  
lty=3, cex=0.6, xaxt="n", col=1)
```

```
axis(1, at=c(1, 10, 15, 26, 32), labels=format(quarter[c(1, 10, 15, 26, 32)]))
```

```
lines(pred_bm_cv_i, lty=2, lwd=2, col=2)
```

```
lines(pres2, lty=1,lwd=2, col=4)
```

```
legend("topleft", legend=c("BM_cv","BM_cv+SARMAX"), lty=c(2,1), lwd = c(2,2), col = c(2,4))
```

```
#
```

```
plot(cumsum(player), type= "b",xlab="Quarter", ylab="Total Change in Player Numbers",  
pch=16, lty=3, cex=0.6, xaxt="n",col=1)
```

```
axis(1, at=c(1, 10, 15, 26, 32), labels=format(quarter[c(1, 10, 15, 26, 32)]))
```

```
lines(pred_bm_cv, lty=2, lwd=2, col=2)
```

```
lines(cumsum(pres2), lty=3, lwd=2, col=4)
```

```
legend("topleft", legend=c("BM_cv","BM_cv+SARMAX"), lty=c(2,3), lwd = c(2,2), col=c(2,4))
```